ABDUR REHMAN

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SUMMARY

Computer Science student with two internship terms of secure processor verification at an Air Force research lab, plus full-stack mobile development and software engineering project experience. Seeking roles in cybersecurity, secure systems, and software engineering.

EDUCATION

Farmingdale State College, SUNY | School of Engineering Technology — Farmingdale, NY **Expected December 2026**

Bachelor of Science in Computer Science | GPA: 3.71 / 4.00 | Dean's List: Fall 2023 – Present

Relevant Coursework: Data Structures and Algorithms, AI & Machine Learning, Computer Architecture & Organization, Software Engineering, Object-Oriented Programming, Data Management

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Assembly (MIPS and custom ISA), TypeScript, SQL

Frameworks & Platforms: React Native, Expo Router, RESTful APIs, JSON, Google Cloud Firestore, Supabase, JavaFX

Hardware & EDA: FPGA Programming, Cadence Xcelium, Cadence Virtuoso ViVa, Waveform Debug

Developer Tools: Git / GitHub / GitLab, Linux, Visual Studio Code, Vim, IntelliJ IDEA, Azure Data Studio

PROFESSIONAL EXPERIENCE

Secure Processor Intern | Air Force Research Laboratory (AFRL) — Rome, NY **June 2025 – Aug. 2025; June 2026 – Present**

- Designed and debugged assembly-level regression tests for a custom secure processor, restructuring a monolithic test suite into **7** standalone regression tests and identifying **2** critical hardware bugs missed by the original test design
- Diagnosed and fixed a silent test-harness failure that was masking false passes across the regression suite
- Traced hardware execution using Cadence Xcelium/ViVa waveform analysis to localize a cache-controller lockup bug to a single function, giving design engineers a precise starting point for the fix
- Partnered with senior hardware design engineers to validate security-feature fixes through waveform-level verification, reproducing known bugs on prior hardware commits to confirm test coverage

AI Security Researcher | Farmingdale State College, SURI Program — Farmingdale, NY **June 2024 – Aug. 2024**

- Researched AI-driven cybersecurity under faculty mentorship, applying machine learning and NLP models to large network-traffic datasets to detect anomalies and classify malicious activity
- Evaluated algorithmic bias and data-privacy risk in ML-based threat detection, documenting fairness trade-offs in deployment
- Presented findings at the SURI and IGNITE Symposia; earned a Microcredential in Research

PROJECT EXPERIENCE

LogiTruck | CSC 325 — Software Engineering **Sept. 2025 – Dec. 2025**

- Led a **4**-person development team, coordinating task delegation, GitHub branching workflow, and integration testing to deliver a JavaFX fleet-management desktop application on schedule
- Architected a layered MVC structure separating UI, business logic, and data access, defining the interfaces each teammate implemented against to keep parallel work from colliding
- Integrated Google Cloud Firestore for cloud-backed storage with real-time sync across fleet, driver, and shipment records
- Built full CRUD functionality on structured JSON data models through a client–server-style data access layer

UrduSeekho | Independent Mobile Application **2026 – Present**

- Built an Urdu language-learning app covering script reading, vocabulary, and pronunciation, in React Native/Expo and TypeScript
- Implemented the SM-2 spaced-repetition algorithm, deriving review-interval quality scores automatically from answer correctness
- Built an audio pipeline of self-recorded pronunciation clips, auto-segmented per item by a Python (pydub) splitting script
- Designed an audio-resolution layer on Supabase storage, with Web Speech API text-to-speech fallback for unrecorded items

Lazy Chef | RAM Hacks Hackathon **April 2025**

- Developed a mobile app in **48 hours** that generates recipes from a limited set of on-hand kitchen ingredients
- Integrated the Google Vision API to identify ingredients from user photos, filtering out background clutter to improve accuracy
- Used the Spoonacular API to deliver step-by-step recipes, letting users add or remove ingredients to adjust suggestions